GAUTAM PANDEY

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PROFESSIONAL SUMMARY

Backend Software Engineer (SDE-2) with 2 years and 4 months of experience architecting, scaling, and optimizing distributed microservices in fintech and healthcare domains. Specialized in **Java, Spring Boot, Hibernate, Python, PostgreSQL, AWS, Docker, Kafka, Redis**, and transactional API banking infrastructure. Proven track record in building Digital Rupee (e₹) corporate banking services, designing loosely coupled REST microservices, wallet fee reservation engines with atomic Lua scripting, asynchronous message processing, and secure third-party KYC verification platforms.

TECHNICAL SKILLS

Languages & Core: Java, Python, C++, JavaScript, SQL, Data Structures & Algorithms (DSA), OOP Design
Frameworks & Backend: Spring Boot, Spring Data JPA, Hibernate, Microservices Architecture, Django, Express.js, RESTful APIs, Node.js
Security & Auth: JWT, HMAC Signature Verification, OTP / PIN Authorization, Payload Encryption, Idempotency
Cloud, DevOps & Tools: AWS (EC2, S3, SQS, RDS), Docker, Git, Maven, Postman, Swagger, CI/CD, IntelliJ IDEA, VS Code
Databases & Messaging: MySQL, PostgreSQL, MongoDB, Redis (Lua Scripting, Caching), Apache Kafka, Event-Driven Architecture

WORK EXPERIENCE

Cotopay **Oct 2026 – Present**
Software Development Engineer II (SDE-2) **Delhi NCR, India**

- **Digital Rupee (e₹) Platform:** Developed backend services for the Digital Rupee (e₹) platform, enhancing corporate banking capabilities.
- **Wallet Management & UTR Verification:** Implemented wallet management and UTR verification workflows, improving payment accuracy and preventing duplicate credits.
- **Transaction Authorization & Security:** Built secure transaction authorization APIs, including PIN management, OTP-based recovery, and account lockout mechanisms.
- **Bulk Transaction Processing:** Developed asynchronous bulk transaction processing workflows using AWS S3 and data validation.
- **Webhook Integrations:** Integrated Calendly webhooks with HMAC signature verification and idempotency checks for reliable lead tracking.
- **Cross-Functional Delivery:** Collaborated with cross-functional teams to define API contracts, validate end-to-end workflows, and ensure reliable backend operations.

Luckpay Solutions **Nov 2025 – Oct 2026**
Software Developer (Backend / API Banking) **Delhi NCR, India**

- **API Banking Platform & Architecture:** Owned full lifecycle backend development for API Banking microservices using Java & Spring Boot across merchant onboarding, verification APIs, and wallet management, handling 10,000+ daily requests and driving over $1.5M in transactional volume.
- **Microservices Synchronization & Async Workflows:** Decoupled synchronous dependencies by restructuring flows with Kafka event streaming and AWS SQS queues; added strict timeouts, retries, and idempotency to cut cross-service latency by 45% and maintain high system resilience.
- **Wallet Processing & Financial Ledger:** Engineered high-concurrency merchant wallet engines with pre-transaction balance validation, explicit transaction state tracking (REQUESTED, SUCCESS, FAILED), row-level DB locking, and manual operational adjustment workflows to eliminate balance discrepancies.
- **Third-Party KYC Integrations & Security:** Integrated multi-provider identity and banking verification APIs (PAN, GSTIN, Penny Drop, IFSC, OTP/SMS, WhatsApp) with end-to-end payload encryption/decryption, token auth, and audit trails, reducing onboarding turnaround by 80%.
- **Production Engineering & AWS Deployment:** Containerized Spring Boot microservices using Docker and deployed on AWS infrastructure (EC2, SQS, RDS), establishing centralized logging and UAT/production monitoring to accelerate release velocity by 30%.
- **Payment Gateway & Payment Links:** Built a redirect-flow payin platform from scratch with merchant-service and IP-whitelist validation, plus a merchant-facing Payment Links feature. Integrated 6+ payment platforms (Cashfree, Razorpay, PayU, Easebuzz, PhonePe, JioPay, Decentro) for payments, QR, and transaction management; delivered recurring mandates (ENACH, UPI AutoPay) and cross-border rails via Tazapay and GlomoPay.
- **Generic URL Shortener Service:** Created a generic URL shortener service used across multiple microservices to send short URLs to merchants; the short link redirects the merchant so they can complete their flow, including the VKYC journey.
- **Complex API Integrations & Consent Flows:** Integrated complex APIs such as VKYC, Credit Analytics, Experian, and Account Aggregator, and designed the entire consent-based flow end to end.

Doxper (Meddo) **Jun 2024 – Oct 2025**
Associate Software Developer **Gurugram, India**

- **Full-Stack Healthcare Module Ownership:** Engineered responsive React.js web modules (Patient Dashboard, Doctor Profile, Appointment Management, Report History) backed by Python/Django REST APIs, serving 500+ daily active medical practitioners.
- **Database Optimization & Redis Caching:** Investigated and resolved stale Redis cache invalidation bugs and optimized complex PostgreSQL queries, lowering database load times by 35% and improving application stability during peak hours.
- **Medical Report Generation Engine:** Built automated report uploading, categorizer, and retrieval flows with dynamic A4-format PDF and Excel exports, eliminating manual document processing overhead for clinical teams.

PROJECTS

Blocked Balance System — Real-Time Fee Reservation Engine **Java · Spring Boot · Redis (Lua Scripting) · Kafka/SQS · MySQL**

- Designed and built a **Redis-based fee reservation (blocked balance) system** to **eliminate revenue leakage** on chargeable APIs (BAV, OCR, eSign) caused by asynchronous fee-deduction failures.
- Wrote **atomic Redis Lua scripts** to block and release fees, guaranteeing **thread-safe, idempotent operations** under concurrent merchant API calls and **preventing race-condition over-spending**.
- Engineered a **dual-key Redis schema** (blocked-total + blocked-txns) with **TTL-based safety nets and a retry scheduler**, enabling accurate available balance checks and **automatic recovery** when downstream services were down.
- Built **cross-service MDR plan caching** (dashboard → Redis → API banking) to dynamically resolve per-API merchant fees used in the blocking and wallet-validation flow.

Manual Fund Credit/Debit — Verification API Wallet Adjustment System **Java · Spring Boot · SQS · MySQL · Event-Driven Architecture**

- Designed an **event-driven, SQS-based architecture** enabling operational teams to manually credit or debit merchant Verification API wallets for corrections, reversals, and approved fee adjustments.
- Built dashboard-side `initiateDebitRequest` and `initiateCreditRequest` APIs with **OTP verification, duplicate-request checks, balance validation, and full audit-trail logging**.
- Kept wallet mutations decoupled and safe by routing all balance changes through a **single owning service (lp-paytech-api)**, preserving **row-level locking, ledger entries, and idempotency guarantees**.
- Implemented **completion-event driven status tracking**, updating `apibanking_txn_log` from **REQUESTED to SUCCESS/FAILED** based on downstream wallet-service confirmation.

EDUCATION

Bachelor of Engineering — Computer Science & Engineering **CGPA: 9.3 / 10.0**
Chitkara University, Baddi, HP